MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 6DP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic focus and deflection, 6-inch diameter, flat face, three beam, post-accelerator.

PIN CONNECTIONS AND DIMENSIONS: See figures 1 and 2

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	Edc	Vdc	Vdc	v	Meg Note 1	Meg Note 2	Vdc	Ratio	ft
Maximum:	6.9	0	1,550	4,500	15,000	600	1.5	1.0	±180	3.0	50,000
Minimum:	5.7	-200	---	2,500	8,000	---	---	---	---	---	---
TEST CONDITIONS: (See note 3)	6.3	Adj	Focus	4,000	12,000	---	---	1.0	---	3.0	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	4.625	Inches
5101	Face tilt	17		---	---	0.080	Inch
Ⓒ 5216	Cathode illumination	-		---	---	---	---
5248	Deflection factor uniformity	11		---	---	---	---
---	Interaction factor	15		---	---	$6x10^{-5}$	Inch/Vdc
5256	Magnetization	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all	Ck	---	7.5	pF
			Grid No. 1 to all	Cg1	---	8.0	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	3.0	pF
			D1 to all	CD1	---	16.0	pF
			D2 to all	CD2	---	16.0	pF
			D3 to all	CD3	---	11.0	pF
			D4 to all	CD4	---	11.0	pF
Ⓒ 1216	Base material insulating quality	-	Zone 5, minimum	---	---	---	---
1027	Temperature cycling	-		---	5	---	Cycles
1141	Pressure	-		---	---	---	---
1026	Low-temperature operation	-		---	-55	---	°C
1026	High-temperature operation	-		---	---	+70	°C
5111	Vibration	-		Width	---	1	mm
1002	Barometric pressure, reduced	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	16	Ib3 = 50 µAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5223	Modulation	-	Ib3 = 50 µAdc	ΔEc1	---	50	Vdc
			Ib3 = 25 uAdc	ΔEc1	---	35	Vdc
5231	Spot position (electrostatic deflection)	-		Displ	---	15	mm
5231	Spot displacement (leakage)	-		Displ	---	10	mm
5241	Grid cutoff voltage	-		Ec1	-75	-124	Vdc
---	Minimum useful scan	12		---	---	---	---
---	Pattern distortion	13, 14		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2						
1301	Heater current	4		If	1.50	1.98	A
5201	Electrode current (anode No. 1)	- 5	Ib3 = 50 µAdc Ec1 = 0	Ib1 Ib1	-10 -50	+5 +5	µAdc µA
5201	Electrode current (cathode)	-	Ib3 = 50 µAdc	Ik	---	300	µAdc
5201	Electrode current (beam)	5	Ec1 = 0	Ib3	350	---	µA
5101	Base alinement (electrostatic types)	-	+102, base key	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	102, gun B	---	-3	+3	Degrees
5101	Side terminal and base alinement	-	Base key	---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
© 5101	Angle between traces	-		---	88	92	Degrees
© 5101	Trace alinement (corresponding traces)	-		---	---	2	Degrees
---	Stray emission	6		---	---	---	---
---	Stray emission flare	7	Ib3 = 25 µAdc	---	---	3	Degrees
5221	Screens (P7 types)	8	Metallized cb5 = 42% cb (min); GS:1 = 3 (min)	---	---	---	---
© 5226	Line width A (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.38	mm
© 5226	Line width B (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.65	mm
© 5246	Focusing voltage at cutoff	10		Eb1	---	1,225	Vdc
5246	Focusing voltage (zero bias)	5, 10		Eb1	950	---	Vdc
© 5246	Focusing voltage	10	Ib3 = 25 µAdc	Eb1	1,000	1,225	Vdc
© 5248	Deflection factor	-	102	DF	112	138	Vdc/inch
© 5248	Deflection factor	-	304	DF	105	130	Vdc/inch
5251	Heater-cathode leakage current	-		---	---	---	---
© 1111	Base pin solder depth	-		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---
© 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group D; Ib3 = 25 µAdc; t = 500 hours (minimum)	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued						
---	Life-test end points:						
5226	Line width A	-	Ib3 = 19 µAdc	Width	---	0.38	mm
5226	Line width B	9	Ib3 = 19 µAdc	Width	---	0.65	mm
5223	Modulation	-	Ib3 = 19 µAdc	ΔEc1	---	35	Vdc
5201	Electrode current (beam)	5	Ec1 = 0	Ib3	200	---	µA

NOTES:

1. Applications requiring high grid No. 1 impedance may use up to 5 megohms under controlled conditions.
2. It is recommended that the deflecting electrode circuit resistances be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. All tests except vibration, capacitance, interaction factor, and environmental tests shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1 megohm.
4. Three heaters paralleled internally.
5. To protect the tube against damage, the accelerator power input (average) shall be limited to 6 watts; zero bias shall be maintained for a period not exceeding 2 seconds, followed by a cutoff period of not less than 10 seconds.
6. With the tube enclosed in a lightproof container, with the specified potentials applied, and with no deflecting fields applied, the tube shall be biased to cutoff. Under these conditions, the entire face of the tube shall be examined for stray emission by an observer who has accommodated his eyes to viewing the face of the cathode-ray tube for at least 2 minutes through an aperture in the lightproof container. If stray emission occurs, scanning fields such as would be required to produce a 1-by 1-inch (25.4 mm) raster, shall be applied, under which condition the stray emission shall disappear completely.
7. A 35- to 105-line pattern shall be provided on the cathode-ray tube screen, using 60-cycle sawtooth scanning on one axis and 2100- to 6300-cycle sawtooth scanning on the other axis. The mean potential of the deflection electrodes shall be that of the accelerator. The pattern size shall be adjusted to 3/8 to 3-1/2 inches (9.53 to 88.90 mm), the high-frequency scanning being in the long direction of the rectangular pattern. The pattern then shall be moved plus or minus 1.625 inches (41.28 mm) from the center line of the face by balanced dc deflection voltages applied to the pair of deflection electrodes which produces the 3/8 inch (9.53 mm) scan. This procedure shall be repeated in the other axis by reversing the deflection electrode connections from 1D2 to 3D4 and vice versa. In the extreme position of the pattern, the light output at the face of the tube produced by stray electrons shall not be visible at any point when viewed at a distance of 6 feet and under low-ambient light conditions, through a pair of Polaroid HN-32 polarizers, or equivalent with the axes of polarizers at the specified angle from perpendicular.
8. The P7 screen light buildup shall be tested at an Eb3 voltage of 12,000 Vdc, Eb2 voltage of 4,000 Vdc and a screen power of 0.24 watt.
9. Measurement shall be made at ±1.625 inches (41.28 mm) from the tube face center.
10. Scanning pattern and focus adjustment as specified in method 5226.
11. The minimum useful scan shall be used in place of minimum useful screen diameter.
12. Useful scan display on face of tube shall be as shown on figure 2.
13. The total vertical movement of the upper or lower ends of a 3-inch (76.20 mm) vertical trace, (centered with respect to the tube face), deflected horizontally 1.5 inches (38.10 mm) to the left and right of the center of the tube face, shall not exceed 0.075 inch (1.91 mm). The 3D4 trace shall be considered vertical.
14. The total horizontal movement of the left or right ends of a 3-inch (76.20 mm) horizontal trace, (centered with respect to the tube face), deflected vertically 1.5 inches (38.10 mm) above and below the center of the tube face, shall not exceed 0.075 inch (1.91 mm). The 1D2 trace shall be considered horizontal.
15. The deflection per unit voltage of one beam at any position of the beam within its useful scan when a balanced voltage of at least ±300 Vdc is applied to the 1D2 electrodes of all of the other electron guns shall be not greater than the value specified.

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NOTES: -Continued

16. This test to be performed at the conclusion of the holding period.

17. The axis of the neck is defined as the axis of rotation when the neck is supported on two sets of V wheels, 3-wheel chucks, or equivalent, placed 0.75 inch (19.05 mm) and 3.0 inches (76.20 mm) below CD line No. 2 (see figure 1).

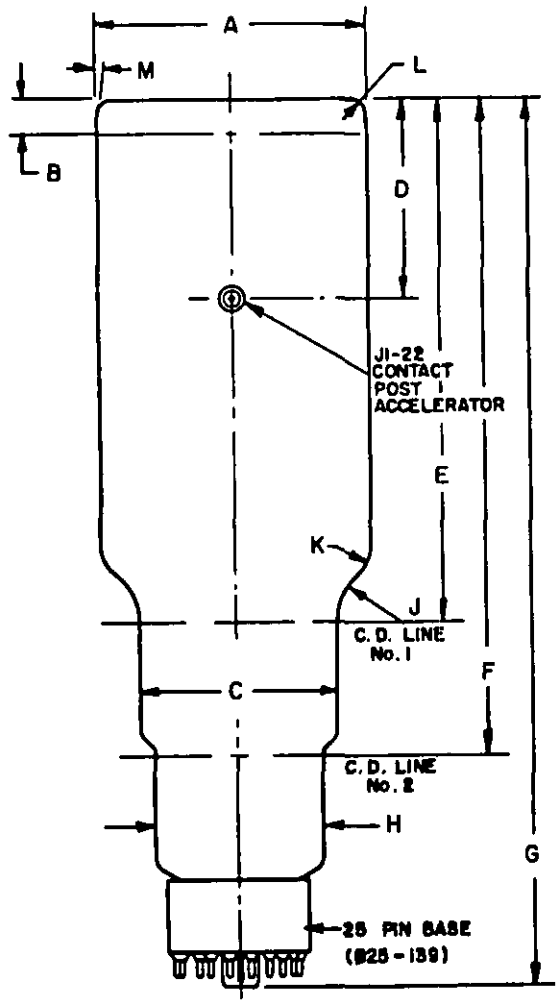
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Navy - AS, OS, MC, CG, SH

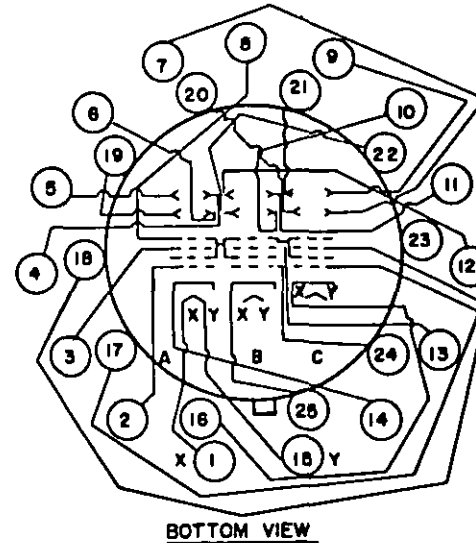
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PIN No.	ELEMENT
	GUN A
14	CATHODE
2	GRID No. 1
3	FOCUSING ELECTRODE
5	DEFLECTOR D1
4	DEFLECTOR D2
6	DEFLECTOR D3
19	DEFLECTOR D4
23	NO CONNECTION
	GUN B
25	CATHODE
24	GRID No. 1
13	FOCUSING ELECTRODE
12	DEFLECTOR D1
11	DEFLECTOR D2
10	DEFLECTOR D3
22	DEFLECTOR D4
	GUN C
16	CATHODE
17	GRID No. 1
18	FOCUSING ELECTRODE
21	DEFLECTOR D1
9	DEFLECTOR D2
7	DEFLECTOR D3
20	DEFLECTOR D4
8	ACCELERATOR (COMMON)
1 AND 15	HEATER (COMMON)

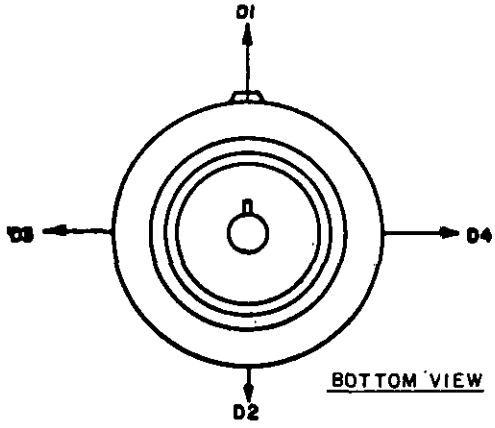


FIGURE 1. Outline drawing of electron tube type 6DP7.

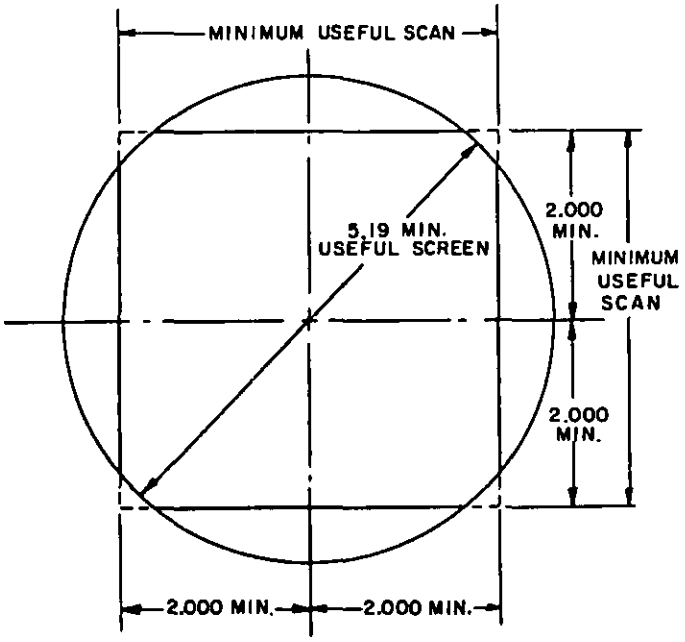
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.750 (146.05)	5.938 (150.83)
B	.750 (19.05)	1.000 (25.40)
C	4.312 (109.52)	4.500 (114.30)
D	4.250 (107.95)	4.750 (120.65)
E	11.500 (292.10)	12.000 (304.80)
F	---	14.000 (355.60)
G	19.687 (500.05)	20.437 (519.10)
H	3.656 (92.86)	3.844 (97.64)
Reference dimensions (see note b)		
J	1.406 (35.71) R	
K	1.000 (25.40) R	
L	.250 (6.35) R	
M	3° to 5°	

NOTES:
a. J1-22 contact to aline with base key ±10°.
b. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 6DP7 - Continued.

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INCHES	MM
2.000	50.80
5.19	131.83

FIGURE 2. Minimum useful scan.

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